# Application of Saddlepoint Approximation in Honing Forecast

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March 2026

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# 1 Introduction

In this document, we will first define what we want to compute, then go through how we do it. This is aimed at an audience who fall in the intersection of Lost Ark players and aspiring mathematicians. As such, we will use rigorous notations to make our statements and logic clear without losing track of our greedy goblin goal of saving gold.

## 2 Setup

We must first endure some boring definitions, after which we will find that the math is rather simple up until the very last part.

### 2.1 Basics

Let $U_j = U_1, U_2, \ldots, U_N$ be the upgrades we want to perform, which we will view as discrete random variables[1]. Note that the upgrades are independent, and take value in $0, 1, \ldots n_j$, representing the number of attempts needed for each upgrade.
Let $X = (U_1, U_2, \ldots, U_N)$ be the random vector representing the overall outcome.
Let $C_i(x)$ be the material cost function for material $i$, where $i \in 1, 2, \ldots, M$. For ordinary mats, this is of the form $C_i(x) = u_i + c_i \cdot x$ for some constants $u_i$ and $c_i$. However, this is arbitrary for juices (but always increasing).
Let $b_i$ be the "budget", the amount of bound material $i$ that we own.
Let $m_i$ and $l_i$ be the market price and leftover value of material $i$.
Let $g_i(x)$ be our gold cost function, defined as follows[2]:

$$g_i(x) = \begin{cases} (x - b_i) \cdot m_i & \text{if } x > b_i \\ (x - b_i) \cdot l_i & \text{otherwise} \end{cases}$$

As in, every unit of material $i$ that we consume beyond our budget incurrs $m_i$ gold cost, and every unit of $i$ that we have leftover is worth $l_i$ gold. By default, $l$ is 0.

As such, we can consider the gold cost due to material $i$:

$$G_i(X) = g_i(\sum_{j=1}^{N} C_i(U_j))$$

Note that $g_i$ is applied to the sum, not the cost of the individual upgrades.

[1] This implicitly assumes that the user will not update their progress by simplifying the multi-step process into a single realization of the RV $X$. As in we answer the question "What is the optimal plan given the information we have, assuming that we cannot change our plan after we start?" We can probably do better if we account for the fact that we will gain information, but I wouldn't even know where to start with that.

[2] With the introduction of tradable mats, this is has 2 breakpoints. The math is the same, except section 4 would be a bit messier.

## 2.2 Special honing

There's an additional layer of complexity we must define:
Let $U_k^*$ be the "skipped" version of the random variable[3].
Let $X'_k = (U_1^*, \cdots, U_k^*, U_{k+1}, \cdots, U_N)$, the overall outcome with $k$ skipped upgrades.
Let $U'_j = U_j^*$ if $j \leq k$, otherwise $U'_j = U_j$. As such $X'_k = (U'_1, \cdots, U'_N)$.
Let $p_k$ be the chance that we skip exactly $k$ upgrades, where $k \in 0, 1, \ldots, N$

## 2.3 What we're computing

To describe this in plain English, this would be "Average overall gold cost incurred due to needed mats". Making use of the notations we've built up, this is:

$$\begin{aligned} S &= \sum_{k=0}^{N} (p_k \cdot \sum_{i=1}^{M} \mathbb{E}[g_i(\sum_{j=1}^{N} C_i(U'_j))]) \\ &= \sum_{k=0}^{N} (p_k \cdot \sum_{i=1}^{M} \mathbb{E}[G_i(X'_k)]) \end{aligned}$$

The word "Average" is doing a lot of heavy lifting here. Note that since expectation is distributive over sums[4], we are allowed to add up the contributions of each material dimension (index $i$ in the sum above).

There are 3 layers of multi-dimension-ness at work here: Special honing, different material types, and different upgrades. The first two are trivial and have been written as sums. However, $G_i$ is a non-linear function that depends on the overall outcome, not just the outcome of a single upgrade, so we cannot factor it out. And so our journey of unwrapping this bundle of joy begins.

[3]This is just a deterministic constant equal to the number of normal honing attempts the user did before doing special hones, usually equal to 0 (but does not necessarily mean $C_i(U'_k) = 0$).

[4]This holds even for non-independent random variables, which the cost dimensions, $G_i(X'_k)$, are not (because they depend on the same underlying RV).

## 3 Special honing: $p_k$

We must recognize that we have made a simplification here - we assume that we must keep doing special hones on one upgrade until either we run out or we succeed. (The alternative being attemping A for a times, then B for b times, then back to A etc etc)

This assumption has a few motives / implications:

1. The decision space of special hones is just the permutations of $U_1, U_2, \ldots, U_N$. This makes the optimizer's job a lot easier, even if this might forbid some better strategies.

2. Succeeding $k$ implies that we succeeeded $k-1$, which has 2 effects:

   (a) This drastically reduces the number of possibilities we need to consider, and therefore the number of times we need to evaluate the inner sum. As in, we only need to consider $(fail, f, f)$, $(success, f, f)$, $(s, s, f)$, $(s, s, s)$ etc, as opposed to the cartesian product $\{s, f\}^n$. This reduces the complexity from $2^n$ to $(n+1)$. This is by far the biggest concern[5].

   (b) This makes the computation of $p_k$ a lot easier. This is done using some dynamic programming.

3. This is easier for the user to follow.

## 4 Contribution from one cost dimension: $\mathbb{E}[G_i(X'_k)]$

Unfortunately, for a non-linear function $f$, $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$, which is the whole reason why we're here. Recall that:

$$G_i(X) = g_i(\sum_{j=1}^{N} C_i(U_j))$$

where $g_i$ is non-linear. If this was an arbitrary function, evaluating this would require either a (brute force) convolution of random variable or something like Fast Fourier Transform(I think). However, we know that our $g_i$ is piecewise linear:

$$g_i(x) = \begin{cases} (x - b_i) \cdot m_i & \text{if } x > b_i \\ (x - b_i) \cdot l_i & \text{otherwise,} \end{cases}$$

[5] Although if we limit our decision space to allow for a limited number of switches, this COULD be viable. However even allowing for 1 switch doubles the amount of SA evaluations, so this would only be vaguely viable for a small number of upgrades.

and so we can do something more clever. For convenience define

$$Y_{ik} = \sum_{j=1}^{N} C_i(U'_j), \text{ so } G_i(X'_k) = g_i(Y_{ik}).$$

So $Y_{ik}$ is the random variable representing the cost of material type $i$ with $k$ skipped upgrades. We can notice that :

$$\mathbb{E}[g_i(Y_{ik})] = m_i \cdot \mathbb{E}[(Y_{ik} - b_i) \cdot I(Y_{ik} > b_i)] + l_i \cdot \mathbb{E}[(Y_{ik} - b_i) \cdot I(Y_{ik} < b_i)],$$

where $I$ is the indicator function. From here, we will drop the subscripts $_i$ and $_k$ as we're focusing on one material type dimension and one special outcome. We can split this into three parts:

$$\begin{aligned}\mathbb{E}[g(Y)] &= m \cdot \mathbb{E}[Y - b] + (m - l) \cdot E[(Y - b) \cdot I(Y < B)] \\ &= m \cdot \underbrace{\mathbb{E}[Y - b]}_{\text{part a}} + (m - l) \cdot (\underbrace{E[Y \cdot I(Y < B)]}_{\text{part b}} - \underbrace{E[b \cdot I(Y < B)]}_{\text{part c}})\end{aligned}$$

Part $a$) is trivial to calculate because $Y$ is a simple sum of random variables, making $\mathbb{E}$ distributive.
For part $b$) and $c$), we will write the expectation in a form that can leverage saddlepoint approximation. We define the biased distribution $Y^*$, which has the probabily distribution $p^* = p \cdot \frac{y}{\mathbb{E}[Y]}$ for each $y$ in the support of $Y$. In other words:

$$\mathbb{P}(Y^* = y) = \frac{y\mathbb{P}(Y = y)}{\mathbb{E}[Y]},$$

such that $Y^*$ satisfies:

$$\mathbb{E}[Y \cdot I(Y < b)] = \mathbb{P}(Y^* < b) \cdot \mathbb{E}[Y].$$

,
Part $c$) is a simpler version of part $b$), we have:

$$\mathbb{E}[b \cdot I(Y < b)] = \mathbb{P}(Y < b) \cdot b.$$

## 4.1 Recap

Recall that we want to compute the following:

$$\begin{aligned}S &= \sum_{k=0}^{N} (p_k \cdot \sum_{i=1}^{M} \mathbb{E}[G_i(X'_k)]) \\ &= \sum_{k=0}^{N} (p_k \cdot \sum_{i=1}^{M} \mathbb{E}[g_i(Y_{ik})]).\end{aligned}$$

With the rearrangements we did above, the inner expectation has been reduced to the following form:

$$\mathbb{E}[g_i(Y_{ik})] = m \cdot \mathbb{E}[Y_{ik} - b] + (m - l) \cdot (\mathbb{P}(Y_{ik}^* < b) \cdot \mathbb{E}[Y_{ik}] - \mathbb{P}(Y_{ik} < b) \cdot b).$$

As such, we have set up our problem such that all we need is some way to compute $\mathbb{P}(Y^* < b)$ and $\mathbb{P}(Y < b)$.

# 5 Probability that we stay below budget: $\mathbb{P}(Y < b)$

We use a procedure known as saddleppoint approximation to perform this computation. I will describe the steps below:

## 5.1 Cumulant generating function $K(\theta)$

We compute the cumulant generating function $K(\theta)$, where $K$ is different for a particular $Y_{ik}$ but we will spare the subscripts. $K(\theta)$ is defined as follows:

$$K(\theta) = \log(\mathbb{E}[e^{\theta Y}]).$$

$K$ has the property that it is distributive over addition. Recall that $Y_i = \sum_{j=1}^{N} C(U_j)$, so we have:

$$K(\theta) = \sum_{j=1}^{N} \log(\mathbb{E}[e^{\theta C(U_j)}]).$$

This property allows us to compute the individual summands and add them up at the end to obtain $K(\theta)$. Furthermore, it is also easy to compute its derivatives $K'(\theta)$, $K''(\theta)$, $K'''(\theta)$, $K^{(4)}(\theta)$ (we stop at the 4th cumulant for floating point reasons).

## 5.2 Find $\theta$ such that $K'(\theta) \approx b^+$

Since we have access to higher derivatives of $K'(\theta)$, we can use a generalized version of newton's algorithm for root finding, known as Householder's method. My particular implementation uses a bunch of heuristics to avoid non-convergence that are complete trial and error, so there may be faster implementations.

### 5.3 $\mathbb{P}(Y < b) =$ magic formula

The specific version of saddlepoint approximation implemented is as follows:

$$\mathbb{P}(Y < b) = \begin{cases} \Phi(w) + \phi(w)(\frac{1}{w} - \frac{1}{u}) & \text{if } x \neq \mu \\ \Phi(z) - \phi(z)\left[\frac{\kappa_3}{6}H_2(z) + \frac{\kappa_4}{24}H_3(z) + \frac{\kappa_3^2}{72}H_5(z)\right] & \text{if } x \approx \mu, \end{cases}$$

$$\begin{aligned} \text{where } \delta &= \text{minimum gap size in the support of Y}, \\ b^+ &= b + \delta/2, \\ \theta &\text{ satisfies } K'(\theta) \approx b^+, \\ w &= \operatorname{sgn}(\theta)\sqrt{2(\theta b^+ - K(\theta))}, \\ u &= 2\delta^{-1}\sinh(\theta\delta/2)\sqrt{K''(\theta)}, \\ z &= (b^+ - \mu)/\sigma, \\ \kappa_n &= K^{(n)}(0)/\sigma^n, \\ H_2(z) &= z^2 - 1, \\ H_3(z) &= z^3 - 3, \\ H_5(z) &= z^5 - 10z^3 + 15z, \\ \text{and } \Phi, \phi &\text{ are the cdf and pdf of Normal distribution.} \end{aligned}$$

The derivation of this formula can be found in Saddlepoint Approximations with Applications by Butler [1]. Specifically, the base form (Lugannani-Rice formula) is stated in Section 1.2.1, the continuity corrected version in Section 2.4.4, and the limiting case in Section 5.2.3.

### 5.4 Cumulants of $Y^*$

Since $Y^*$ is not a sum of random variables, $K$ is not distributive over its summands, so naively computing the cumulant generating function of $Y^*$ would again require a direct convolution. Fortunately, It turns out the cumulant of $Y^*$ and $Y$ are related by the following:

$$K_{Y^*}(\theta) = K_Y(\theta) + \log(K_Y'(\theta)) - \log(\mu),$$

where higher cumulants can similarly be found via simple formulae.

## 6 Conclusion

To recap, we did the following:

1. For each material, compute the probability that we exceed our budget (Section 5)
2. Using this, we compute the average gold contribution due to this mat (Section 4)
3. We add this up for all mats. This is the contribution of one special honing scenario.
4. We then take the weighted average for each free tap scenario (Section 3).

## References

[1] Ronald W Butler. *Saddlepoint approximations with applications*, volume 22. Cambridge University Press, 2007.